Group 4 - Hybrid Approach for Optimization

Objective:

Develop a **hybrid object detection model** that leverages the **speed of YOLO** for region proposals and the **accuracy of Faster R-CNN** for final classification. The goal is to evaluate whether this hybrid approach achieves a better trade-off between detection accuracy and inference speed compared to standalone models.

Task Breakdown:

1. Understanding the Hybrid Approach

- YOLO is extremely fast but may struggle with fine-grained object localization.
- Faster R-CNN is highly accurate but computationally expensive.
- The **hybrid approach** aims to combine the strengths of both:
    1. **YOLO for region proposal**: Use YOLO to generate potential bounding boxes quickly.
    2. **Faster R-CNN for refinement**: Use Faster R-CNN to classify objects within these bounding boxes.

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2. Implementation Steps

A. YOLO-Based Region Proposal

- Use **YOLOv5 or YOLOv8** to detect objects in an image.
- Instead of using YOLO's classification, extract only the **bounding boxes** with high confidence scores (e.g., threshold > 0.5).
- Pass these bounding boxes as **region proposals** to the next stage.

B. Faster R-CNN for Classification & Refinement

- Crop the detected regions from YOLO's output and **feed them into a pre-trained Faster R-CNN**.
- Allow Faster R-CNN to **refine bounding boxes and classify objects**.
- The final output should display the **optimized bounding box and label**.

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3. Performance Evaluation

A. Mean Average Precision (mAP)

- Compare **mAP@0.5** and **mAP@0.5:0.95** for:
    - **Standalone YOLO**
    - **Standalone Faster R-CNN**
    - **Hybrid YOLO + Faster R-CNN**

B. Inference Time (Speed)

- Measure the **inference time per image** for all three methods.
- Compute the **speedup ratio** of the hybrid approach compared to Faster R-CNN alone.

C. Detection Accuracy & False Positives/Negatives

- Compare the **number of false positives and false negatives** across models.
- Analyze whether the hybrid model improves detection precision.

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4. Final Report & Analysis

Each group must submit a report covering:

1. **Architecture of the hybrid model** – How YOLO and Faster R-CNN are integrated.
2. **Performance comparison** – mAP, inference time, accuracy, and bounding box quality.
3. **Trade-off analysis** – Is the hybrid approach better than standalone models?
4. **Graphical analysis**:
    - Performance comparison plots.
    - Sample detections showing improvements with hybrid optimization.
5. **Discussion on real-world applications**:
    - **Does this approach benefit real-time systems?**
    - **Can this hybrid technique improve performance in applications like surveillance or autonomous driving?**

Deliverables

✓ **Code Implementation**: Hybrid object detection pipeline using YOLO & Faster R-CNN.
✓ **Performance Metrics**: mAP, inference time, and accuracy comparison.
✓ **Comparison Report**: Analysis of hybrid vs. standalone models.
✓ **Visualizations**: Sample detections and bounding box comparisons.